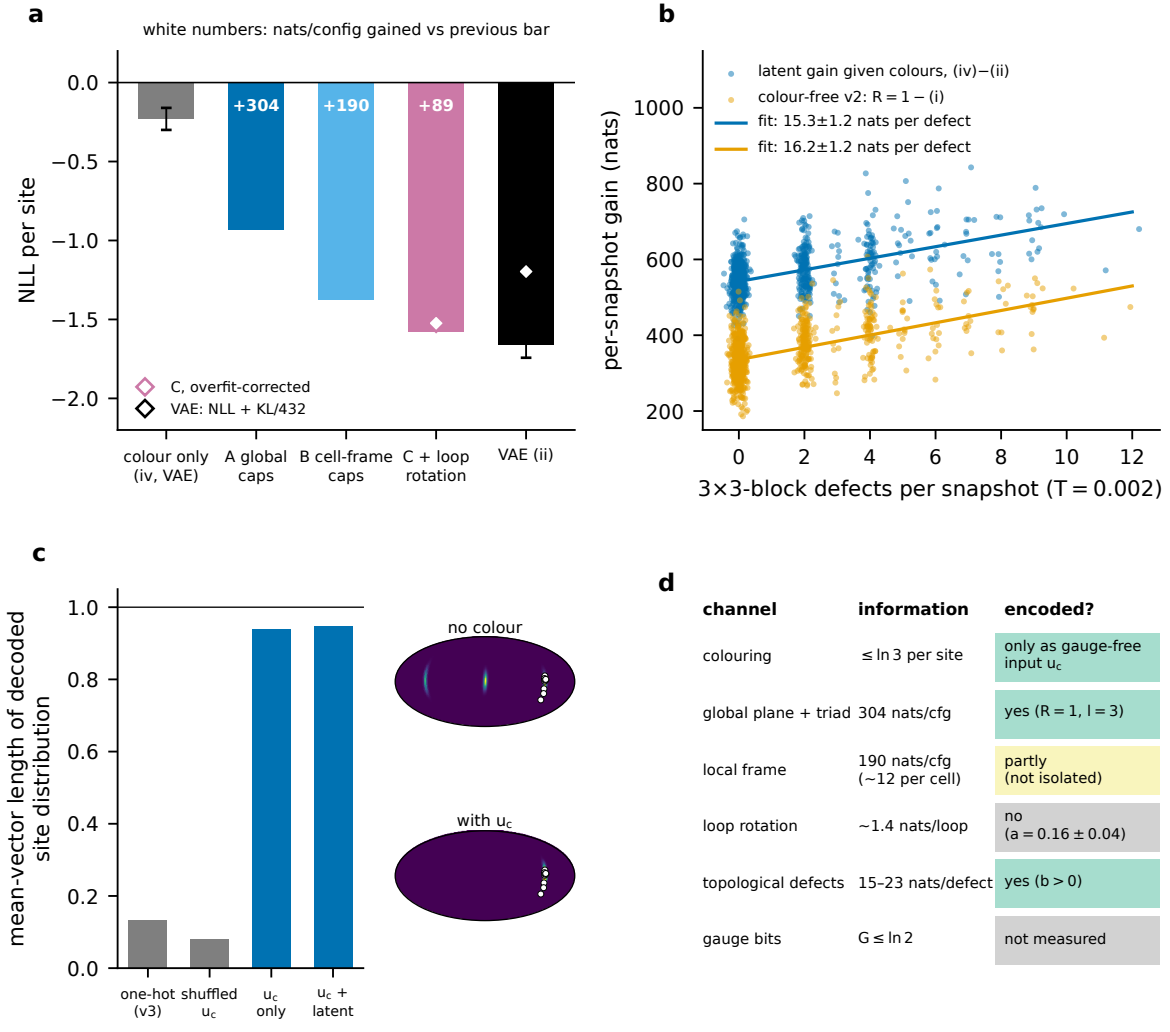

a
white numbers: nats/config gained vs previous bar
NLL per site
0.0
−0.5
−1.0
−1.5
−2.0
+304
+190
+89
C, overfit-corrected
VAE: NLL + KL/432
colour only (iv, VAE)
A global caps
B cell-frame caps
C + loop rotation
VAE (ii)
b
latent gain given colours, (iv)−(ii)
colour-free v2: R = 1 − (i)
fit: 15.3±1.2 nats per defect
fit: 16.2±1.2 nats per defect
per-snapshot gain (nats)
1000
800
600
400
200
0
2
4
6
8
10
12
3×3-block defects per snapshot (T = 0.002)
c
mean-vector length of decoded site distribution
1.0
0.8
0.6
0.4
0.2
0.0
one-hot (v3)
shuffled u_c
u_c only
u_c + latent
no colour
with u_c
d
channel
information
encoded?
colouring
≤ ln 3 per site
only as gauge-free input u_c
global plane + triad
304 nats/cfg
yes (R = 1, I = 3)
local frame
190 nats/cfg (~12 per cell)
partly (not isolated)
loop rotation
~1.4 nats/loop
no (a = 0.16 ± 0.04)
topological defects
15–23 nats/defect
yes (b > 0)
gauge bits
G ≤ ln 2
not measured